

RDXXB SERIES RADAR

External Communication Protocol

X-Band · Mechanically Scanned · Drone Detection

Version	V1.0
Release Date	21 April 2026
Scope	RDXXB Full Series
Protocol	UDP · Little-Endian
Classification	Technical Reference

1. Document Overview

This document specifies the external communication protocol for the RDXXB series of mechanically scanned radars and their associated display/control systems. The protocol defines the complete set of UDP frame formats for command, status, and target data exchange between a radar front-end and a display/control application.

1.1 Version History

[illegible]

2. Communication Protocol

All communication between the radar and the display/control system uses the UDP network protocol. All multi-byte fields are encoded in little-endian order (low byte first, high byte last).

2.1 Network Configuration

Endpoint	IP Address	Port	Direction
Display / Control System	192.168.1.2	8000	Receives data; sends commands
Radar Unit	192.168.1.3	7000	Sends target/status data; receives commands
Broadcast (Discovery)	255.255.255.255	11111	Control sends arbitrary content; radar replies with IP and port

2.2 Byte Order

All data fields wider than one byte are transmitted in little-endian byte order: the least-significant byte (low byte) is placed first, the most-significant byte (high byte) last.

■ **Note:** Example: the value 0x1234 is transmitted as bytes [0x34, 0x12] on the wire.

2.3 Frame Identifier Map

Frame ID	Frame Name	Direction	Notes
0xFF00	Mode Command Frame	Display → Radar	Controls operating mode
0xFF01	Tracking Target Information Frame	Radar → Display	Per-pulse, variable length
0xFF02	Search Target Information Frame	Radar → Display	Per-pulse, variable length
0xFF03	Status Information Frame	Radar → Display	~200 ms period
0xFF04	Parameter Update Frame	Display → Radar	In search mode only
0xFF05	Scan Mode Switching Command Frame	Display → Radar	Switches scan strategy
0xFF30	RESERVED	–	Do NOT transmit
0xFF31	RESERVED	–	Do NOT transmit
0xFF32	RESERVED	–	Do NOT transmit
0xFF33	Network Configuration Frame	Bidirectional	Discovery / response
0xFF34	RESERVED	–	Do NOT transmit

3. Data Protocol

3.1 Universal Frame Format

Every communication frame—regardless of type—shares the following outer wrapper. The frame content is the only variable-length portion; its size in bytes is reported by `content_length`.

Field Name	Bytes	Data Type	Value / Description
<code>frame_header</code>	2	uint16 (LE)	Always 0x55AA — magic bytes to identify a valid frame
<code>frame_id</code>	2	uint16 (LE)	Frame type identifier (see Section 2.3)
<code>frame_count</code>	4	uint32 (LE)	Monotonic counter 0x00000000 – 0xFFFFFFFF; wraps around; receiver-synchronized
<code>content_length</code>	4	uint32 (LE)	Byte size of the <code>frame_content</code> field (n)
<code>protocol_id</code>	1	uint8	Radar device ID: 1=RD01, 2=RD02, ... 8=RD08. Frames with an unrecognized ID must be silently ignored
<code>protocol_major</code>	1	uint8	Major version (default: 2). A higher major version means a newer, potentially incompatible protocol. New software reads old protocols; old software is NOT guaranteed to read new protocols.
<code>protocol_minor</code>	1	uint8	Minor version (default: 13 for V2.13/V2.14). Within the same major version, both forward and backward compatibility are guaranteed.
<code>checksum</code>	1	uint8	uint8 accumulation of all bytes EXCEPT the 2-byte <code>frame_header</code> . Take the least-significant byte of the sum.
<code>frame_content</code>	n	bytes	Frame-type-specific payload. Length is given by <code>content_length</code> .

■ **Note:** Total frame size = 16 + n bytes.

3.2 Frame Content

3.2.1 Mode Command Frame (0xFF00)

Sent by the display/control system to set or change the radar's operating mode and configuration. The radar defaults to Standby mode on power-on and does NOT respond to duplicate commands. Three modes exist: Standby, Search, and Track. A Leveling mode is also available for pre-operation platform calibration.

Operating Mode Codes

Code	Mode	Description
0x00	Standby	Default on power-on. Radar is powered but not scanning.
0x11	Search	360° scan mode. Radar sweeps and reports detected targets per pulse group.
0x22	Track	Closed-loop tracking of a designated target.
0x33	Leveling	Platform auto-leveling. Set servo to circular scan mode in the command. Do not perform other operations during

		leveling. When status frame reports mode 0x00 or 0x33, leveling is complete.
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Standby / Search Mode Frame Fields

Field Name	Bytes	Data Type	Definition / Values
mode_type	1	uint8	0x00 = Standby 0x11 = Search 0x33 = Leveling
spare	4	char[4]	Reserved, internal use only, fill 0
frequency	1	uint8	Frequency code 0–20 (see Appendix 1). Default: 10 → 9.50 GHz
silent_zone_1_start	2	uint16	0–3600 ($\times 0.1^\circ$). 0x7FFF = invalid/disabled. e.g. 1805 = 180.5°
silent_zone_1_end	2	uint16	0–3600 ($\times 0.1^\circ$). 0x7FFF = invalid/disabled
silent_zone_2_start	2	uint16	0–3600 ($\times 0.1^\circ$). 0x7FFF = invalid/disabled
silent_zone_2_end	2	uint16	0–3600 ($\times 0.1^\circ$). 0x7FFF = invalid/disabled
dist_compensation	2	short	Distance compensation offset. Precision 0.1. Default 0. e.g. 2.3 m → send 23
az_compensation	2	short	Azimuth compensation offset in degrees. Precision 0.1°. Default 0
pitch_compensation	2	short	Pitch compensation offset in degrees. Precision 0.1°. Default 0
scan_cycle	2	short	Scan cycle in 0.1 s increments. Default 32 = 3.2 s. Valid in both Standby and Search.
servo_mode	1	uint8	0x11=Pointing 0x33=Circular Scan 0x77=Pull locking pin 0x88=Insert pin 0xFF=Idle (default in Standby and Search)
servo_pointing_angle	2	uint16	Target angle 0–3600 ($\times 0.1^\circ$) for pointing mode only. e.g. 180.5° → 1805
sys_control_flag	1	uint8	0x00 = do NOT execute system control code 0x01 = execute system control code. Standby only.
sys_control_code	1	uint8	0x01 = power off radar. All others = no action. Standby only.
manual_pitch_enable	1	char	0x00 = Disabled (use leveling data) 0x01 = Enabled (use array-mounted pitch angle instead)
spare2	8	char[8]	Reserved, fill 0
longitude	8	double	Installation site GPS longitude. 6 decimal places.
latitude	8	double	Installation site GPS latitude. 6 decimal places.
altitude	8	double	Installation site elevation. 6 decimal places.
altitude_range	1	uint8	0x11 = 7.5 km range 0x22 = 15 km range. Default: 0x11
spare_config	256	char[256]	Reserved configuration block
height_min_near	2	short	Min height filter for targets ≤ 3 km. Precision 1 m. Default 0 m
height_min_far	2	short	Min height filter for targets > 3 km. Precision 1 m. Default 100 m
height_max	2	short	Max height filter; remove targets at or above this. Precision 1 m. Default 600 m

speed_min	2	uint16	Min absolute speed filter (×0.1 m/s). Default 0
speed_max	2	uint16	Max absolute speed filter (×0.1 m/s). Default 600 = 60 m/s. e.g. 12.3 m/s → 123
spare3	2	uint8	Reserved
range_max	2	uint16	Remove targets further than this distance (m). Default 7500 m
range_min	2	uint16	Remove targets closer than this distance (m). Default 200 m
spare4	30	uint8[30]	Reserved
roll_angle_install	4	float	Array-installed roll angle (2 decimal places). Positive = left-low-right-high viewed along radial. Default 0.0
pitch_angle_install	4	float	Array-installed pitch angle (2 decimal places). Positive = front-low-back-high viewed along radial. Default 0.0
north_angle	4	float	North correction angle. Range [0, 360]. Default 0.0. If GPS heading is valid, recalculated by Appendix II algorithm.
auto_id_enable	1	char	0 = Disabled (ID only via track path) 1 = Enabled (also identifies from search; longer search period)
auto_id_max_targets	1	char	Max targets for autonomous ID. Range: 1–2. Default 2
spare5	30	char[30]	Reserved

Tracking Mode Frame Fields

When the mode type is set to Track (0x22), two additional fields replace the standard body:

Field Name	Bytes	Data Type	Definition / Values
mode_type	1	uint8	0x22 = Tracking mode
track_action	1	uint8	0x11 = Enter tracking 0x22 = Cancel tracking
target_count	2	uint16	Fixed value: 1 (number of targets being tracked)
spare	128	char[128]	Reserved

3.2.2 Status Information Frame (0xFF03)

The radar transmits this frame approximately every 200 ms (non-strict interval). It contains comprehensive health, configuration, and navigation telemetry. Clients may selectively display fields as needed.

Field Name	Bytes	Data Type	Definition / Values
work_mode	1	uint8	0x00=Standby 0x11=Search 0x22=Track 0x33=Leveling
cmd_exec_status	1	uint8	0x00=Idle 0x11=Executing 0x22=Success 0x33=Failed
fault_type	1	uint8	0x00=None 0x01=Parse fail 0x02=Cannot execute Others=custom
radar_model	1	uint8	0x01–0x10 → RD01–RD10
frontend_count_id	1	uint8	Number of front-ends and ID. Fixed: 0x01
frontend1_net_status	1	uint8	0=Not connected 1=Connected
frontend1_scan_cycle	4	uint32	Current scan cycle value
frontend1_azimuth	4	float	[0, 360] degrees
frontend1_pitch	4	float	[0, 360] degrees
frontend1_frequency	2	uint16	0–20; 0=9.2 GHz, 20=9.8 GHz (see Appendix 1)
spare_fe	1	uint8	Reserved
silent_zone1_start	2	short	0–3600 (×0.1°), 0x7FFF = invalid
silent_zone1_end	2	short	0–3600 (×0.1°), 0x7FFF = invalid
silent_zone2_start	2	short	0–3600 (×0.1°), 0x7FFF = invalid
silent_zone2_end	2	short	0–3600 (×0.1°), 0x7FFF = invalid
spare_sz	16	char[16]	Reserved
fan_status	1	uint8	Bits[1:0]=Fan1, [3:2]=Fan2, [5:4]=Fan3, [7:6]=Fan4. 0=Fault, 1=Normal
freq_synth_status	1	uint8	Bit0=LO1, Bit1=LO2. 0=Fault, 1=Normal

3.2.3 Search Target Information Frame (0xFF02)

The radar transmits this frame every pulse group — even when no targets are detected. In that case, only the header is sent with `target_count` = 0. The frame is variable-length: total body size = $N \times 148$ bytes where N = `target_count`.

Search Frame Header (20 bytes, fixed)

Field Name	Bytes	Data Type	Definition
<code>search_azimuth</code>	4	float	Current scan azimuth angle (degrees)
<code>search_elevation</code>	4	float	Current scan elevation angle (degrees)
<code>scan_cycle</code>	4	uint32	Scan cycle counter
<code>pulse_group_id</code>	4	uint32	Current pulse group sequence number
<code>target_count</code>	2	uint16	Number of target records N in body
<code>spare</code>	2	uint8	Always 0x0000

Search Frame Body — Per-Target Record (148 bytes each)

Field Name	Bytes	Data Type	Definition
<code>track_delete_flag</code>	2	uint16	0x00 = Active track 0x01 = Track deleted
<code>track_id</code>	2	uint16	Track ID number: 0 – 10000 (tentative maximum)
<code>loss_count</code>	2	uint16	Consecutive loss count in CPI units
<code>loss_reason</code>	2	uint16	Reason code for track loss
<code>update_time</code>	4	uint32	Track update timestamp (pulse group units)
<code>energy</code>	4	float	Received signal energy (32-bit IEEE 754)
<code>credit_ratio</code>	4	float	Track quality / credit ratio (32-bit IEEE 754)
<code>speed</code>	4	float	Radial speed (m/s, 32-bit IEEE 754)
<code>distance</code>	4	float	Slant range to target (m, 32-bit IEEE 754)
<code>azimuth</code>	4	float	Target azimuth angle (degrees, 0–360, 32-bit IEEE 754)
<code>pitch</code>	4	float	Target elevation/pitch angle (degrees, 32-bit IEEE 754)
<code>height</code>	4	float	Target height above reference (m, 32-bit IEEE 754)
<code>track_type</code>	1	uint8	Target class code (see Section 4: Target Classification)
<code>attributes</code>	1	uint8	Bit-field: search and/or track association flags
<code>radar_id</code>	1	uint8	Radar front-end ID (1–8)
<code>spare1</code>	3	uint8[3]	Reserved
<code>track_point_count</code>	2	uint16	Number of historical track points accumulated
<code>longitude</code>	8	double	WGS-84 target longitude. 6 decimal places.
<code>latitude</code>	8	double	WGS-84 target latitude. 6 decimal places.
<code>altitude_gps</code>	8	double	GPS-derived target altitude (m). 6 decimal places.
<code>vx</code>	4	float	Velocity East component (m/s)
<code>vy</code>	4	float	Velocity North component (m/s)

vz	4	float	Velocity Up component (m/s)
spare_vel	22	uint8[22]	Reserved
envelope_point_count	1	uint8	Number of envelope boundary points
spare_env	13	uint8[13]	Reserved
utc_year	2	short	UTC year (e.g. 2025)
utc_month	1	char	UTC month (1–12)
utc_day	1	char	UTC day (1–31)
utc_hour	1	char	UTC hour (0–23)
utc_minute	1	char	UTC minute (0–59)
utc_second	1	char	UTC second (0–59)
utc_millisecond	1	char	UTC millisecond: 0–99 → 0–990 ms (×10 ms precision)
spare2	19	uint8[19]	Reserved
confidence	1	uint8	Target confidence score. Precision 0.01. Value 87 → confidence 0.87

3.2.4 Tracking Target Information Frame (0xFF01)

Identical structure to the Search Target frame (Section 3.2.3), with the following differences: in mechanical scanning mode, target_count in the header is fixed at 1. The frame is still sent every pulse group even with no target, in which case the body is empty.

■ **Note:** The per-target body record (148 bytes) is identical to the Search frame body. Refer to Section 3.2.3 for field definitions.

Tracking Frame Header (20 bytes, fixed)

Field Name	Bytes	Data Type	Definition
search_azimuth	4	float	Current scan azimuth angle (degrees)
search_elevation	4	float	Current scan elevation angle (degrees)
scan_cycle	4	uint32	Scan cycle counter
pulse_group_id	4	uint32	Current pulse group sequence number
target_count	2	uint16	Number of targets — fixed at 1 in mechanical scanning mode
spare	2	uint8	Always 0x0000

3.2.5 Parameter Update Frame (0xFF04)

Used to dynamically update detection thresholds while the radar is in Search mode. The radar applies these settings to its track management on the next update cycle. This frame may be sent repeatedly.

Field Name	Bytes	Data Type	Definition
spare_header	317	char[317]	Reserved header block
height_min_near	2	short	Min height filter for targets ≤3 km zone (precision 1 m)
height_min_far	2	short	Min height filter for targets >3 km zone (precision 1 m)
height_max	2	short	Max height — remove targets at or above (precision 1 m)
speed_min	2	uint16	Min absolute speed filter (×0.1 m/s)
speed_max	2	uint16	Max absolute speed filter (×0.1 m/s). e.g. 12.3 m/s → send 123
spare_filt	2	uint8	Reserved
range_max	2	uint16	Remove targets further than this distance (precision 1 m)
range_min	2	uint16	Remove targets closer than this distance (precision 1 m)
spare_end	74	char[74]	Reserved

3.2.6 Scan Mode Switching Command Frame (0xFF05)

The display/control sends this frame to switch the radar's scanning strategy. The data processing software safely exits the current mode before entering the new one. Duplicate commands are ignored. The servo can only be controlled in mechanical scanning mode — point the servo to the desired angle BEFORE switching to phase scan mode.

Field Name	Bytes	Data Type	Definition
scan_mode	1	uint8	0x01 = Enter mechanical scanning 0x02 = Enter phase scanning 0x03 = Enter composite (backup). Others treated as mechanical.
spare	15	uint8[15]	Reserved

■ **Note:** In phase scan mode, GPS and inclinometer data cannot be obtained from the current servo angle by default.

4. Target Classification

The `track_type` field in every target record uses the following codes to identify the detected object. Autonomous identification (enabled via `auto_id_enable` in the Mode Command) uses flight-path analysis and, when enabled, search-phase pattern recognition.

Code (hex)	Label	Description
0x00	Unknown	Classification not yet determined or unavailable
0x11	Person	Human on foot
0x12	Car	Ground vehicle
0x20	Drone	Unmanned aerial vehicle (UAV/UAS)
0x24	Bird	Avian – bird or flock
0x31	Vessel	Surface watercraft
0x40	Other	Airborne or ground object not in other categories
0x41	False Target	Detected echo determined to be a false alarm or clutter

5. Coordinate System and Unit Reference

5.1 Angular Encoding (Integer Fields)

Azimuth and elevation angles in command frames are sent as 16-bit unsigned integers with a resolution of 0.1 degrees. To convert:

$$\text{angle_degrees} = \text{raw_integer_value} \times 0.1$$

Example: raw value 1805 → 180.5°. Raw value 3600 → 360.0°.

Invalid azimuth/elevation is marked with the sentinel value 0x7FFF.

5.2 Speed Encoding (Integer Fields in Command/Update Frames)

$$\text{speed_ms} = \text{raw_integer_value} \times 0.1 \quad [\text{m/s}]$$

Example: raw value 123 → 12.3 m/s.

5.3 UTC Timestamp Milliseconds

The `utc_millisecond` field stores values 0–99, representing 0–990 ms at 10 ms precision.

$$\text{actual_ms} = \text{utc_millisecond} \times 10 \quad [\text{ms}]$$

5.4 Confidence Field

The confidence field in target records uses integer representation at 0.01 precision:

$$\text{confidence_ratio} = \text{confidence_byte} \times 0.01$$

Example: value 87 → confidence ratio 0.87.

5.5 Geodetic Coordinates

Longitude, latitude, and altitude in target records are IEEE 754 64-bit doubles (WGS-84 datum), reserved to 6 decimal places of precision. Velocity components V_x (East), V_y (North), and V_z (Up) use the ENU (East-North-Up) reference frame in units of m/s.

5.6 Inclinator Axis Convention

Roll and pitch angles use the following sign convention:

Field	Axis	Positive direction	Negative direction
<code>roll_angle</code>	X (inclinometer)	Left-low, right-high (viewed along radial)	Left-high, right-low
<code>pitch_angle</code>	Y (inclinometer)	Front-low, back-high (viewed along radial)	Front-high, back-low

6. Comprehensive JSON Tracking Output (All Available Data)

The following JSON represents a complete decoded Search Target Information Frame (0xFF02) with ALL available fields from both the Status Information Frame (radar device data) and the target records. This is the COMPLETE dataset available from the RDXB protocol. Select only the fields you need for your application.

```
{
  "frame_metadata": {
    "frame_header": "0x55AA",
    "frame_id": "0xFF02",
    "frame_type": "search_target_information",
    "frame_count": 84231,
    "frame_timestamp_utc": "2025-12-17T08:34:51.820Z"
  },

  "protocol": {
    "id": 1,
    "device_code": "RD01",
    "major_version": 2,
    "minor_version": 14,
    "checksum_valid": true
  },

  "radar_device": {
    "device_id": 1,
    "device_label": "Front-End Radar 1",
    "model_code": "0x01",
    "model_name": "RD01",
    "network": {
      "ip_address": "192.168.1.3",
      "port": 7000,
      "status": "connected",
      "frontend_network_status": 1
    },
  },
  "gps": {
    "longitude": 107.608234,
    "latitude": -6.897456,
    "elevation_m": 150.0,
    "heading_angle_deg": 45.0,
    "heading_valid": true,
    "satellite_count": 12,
    "installed_location_name": "Bandung, West Java, Indonesia"
  },
  "orientation": {
    "roll_angle_deg": 0.5,
    "pitch_angle_deg": 1.2,
    "north_correction_angle_deg": 8.5,
    "inclinometer_z": 0.1
  },
  "operational_state": {
    "work_mode": "0x11",
    "work_mode_label": "search",
    "cmd_execution_status": "0x22",
    "cmd_execution_label": "success",
    "fault_type": "0x00",
    "fault_label": "none",
    "servo_mode": "0x33",
    "servo_mode_label": "circular_scan",
  }
}
```

```

    "scan_mode": "0x01",
    "scan_mode_label": "mechanical_scanning"
  },
  "servo": {
    "status_bits": {
      "encoder_communication_ok": true,
      "driver_communication_ok": true,
      "instruction_timeout": false,
      "command_valid": true,
      "pin_inserted": true,
      "pin_pulled_out": true
    },
    "current_azimuth_deg": 142.3,
    "current_pitch_deg": 5.1,
    "current_scan_cycle": 47
  },
  "frequency": {
    "current_code": 10,
    "current_x_band_ghz": 9.5,
    "current_ku_band_ghz": 16.2,
    "synthesis_status": "0x03",
    "synthesis_status_label": "normal"
  },
  "silent_zones": [
    {
      "zone_number": 1,
      "start_angle_deg": 180.0,
      "end_angle_deg": 200.0,
      "enabled": true
    },
    {
      "zone_number": 2,
      "start_angle_deg": 0.0,
      "end_angle_deg": 0.0,
      "enabled": false
    }
  ],
  "filters": {
    "height_min_near_zone_m": 50,
    "height_min_far_zone_m": 100,
    "height_max_m": 600,
    "speed_min_ms": 0.0,
    "speed_max_ms": 60.0,
    "range_min_m": 200,
    "range_max_m": 7500,
    "altitude_range_mode": "0x11",
    "altitude_range_km": 7.5
  },
  "autonomous_identification": {
    "enabled": true,
    "max_targets": 2
  },
  "thermals": {
    "subarray": [
      { "index": 1, "temperature_c": 45 },
      { "index": 2, "temperature_c": 44 },
      { "index": 3, "temperature_c": 46 },
      { "index": 4, "temperature_c": 45 },
      { "index": 5, "wave_controller_version": "1.2" },
      { "index": 6, "spare": 0 },
      { "index": 7, "spare": 0 },
      { "index": 8, "spare": 0 }
    ]
  }
}

```

```

    ],
    "signal_board_temperature_c": 52,
    "frequency_synthesizer_temperature_c": 48.5,
    "frequency_synthesizer_id": "FS-001"
  },
  "power": {
    "subarray_currents_a": [
      { "index": 1, "current_a": 5.4 },
      { "index": 2, "current_a": 5.3 },
      { "index": 3, "current_a": 5.5 },
      { "index": 4, "current_a": 5.4 },
      { "index": 5, "current_a": 2.1 },
      { "index": 6, "current_a": 2.0 },
      { "index": 7, "current_a": 2.1 },
      { "index": 8, "current_a": 2.0 }
    ],
    "signal_voltages_v": [
      { "channel": 1, "voltage_v": 12 },
      { "channel": 2, "voltage_v": 5 },
      { "channel": 3, "voltage_v": 3.3 }
    ]
  },
  "system_health": {
    "fan_status": {
      "fan1": "normal",
      "fan2": "normal",
      "fan3": "normal",
      "fan4": "normal"
    },
    "memory_status": {
      "qdr_initialized": true,
      "ddr_0_initialized": true,
      "ddr_1_initialized": true,
      "pcie_link_ok": true
    },
    "ad_channels": {
      "sum_channel_over_range": false,
      "pitch_channel_over_range": false,
      "azimuth_channel_over_range": false,
      "backup_channel_over_range": false
    }
  },
  "firmware": {
    "fpga_version": "v3.21",
    "data_processing_version": "2.14.05.2025",
    "pitch_frame_angle": 0.0,
    "pitch_adjustment_level": 0.0
  }
},

"scan_header": {
  "search_azimuth_deg": 142.3,
  "search_elevation_deg": 5.1,
  "scan_cycle_count": 47,
  "pulse_group_id": 113842,
  "target_count": 2
},

"targets": [
  {
    "track_identification": {
      "track_id": 1042,

```

```
    "track_delete_flag": "0x00",
    "track_status": "active",
    "loss_count_cpi": 0,
    "loss_reason_code": 0,
    "update_time_cpi": 113842,
    "age_pulses": 0
  },
  "classification": {
    "track_type_code": "0x20",
    "track_type_label": "drone",
    "attributes_bitfield": "0x01",
    "radar_frontend_id": 1,
    "confidence": 0.87,
    "confidence_pct": "87%"
  },
  "signal_quality": {
    "energy": 2847.43,
    "energy_normalized": 0.92,
    "credit_ratio": 0.92,
    "credit_ratio_pct": "92%"
  },
  "polar_coordinates": {
    "distance_m": 1243.7,
    "azimuth_deg": 142.3,
    "elevation_pitch_deg": 7.2,
    "height_m": 155.8,
    "radial_speed_ms": -3.4,
    "radial_speed_label": "approaching"
  },
  "geodetic_coordinates": {
    "longitude": 107.612483,
    "latitude": -6.903124,
    "altitude_m": 156.0,
    "distance_from_radar_m": 1243.7,
    "bearing_from_radar_deg": 142.3,
    "altitude_above_radar_m": 6.0
  },
  "velocity_enu": {
    "vx_east_ms": -2.1,
    "vy_north_ms": -2.7,
    "vz_up_ms": 0.3,
    "total_speed_ms": 3.42,
    "horizontal_speed_ms": 3.41,
    "vertical_direction": "ascending"
  },
  "track_history": {
    "track_point_count": 38,
    "envelope_point_count": 6,
    "track_age_pulses": 38,
    "track_confidence_accumulated": true
  },
  "utc_timestamp": {
    "year": 2025,
    "month": 12,
    "day": 17,
    "hour": 8,
    "minute": 34,
    "second": 51,
    "millisecond": 820,
    "millisecond_precision_10ms": 82,
    "iso8601": "2025-12-17T08:34:51.820Z",
    "unix_timestamp": 1766049291.82
  }
}
```

```

    }
  },
  {
    "track_identification": {
      "track_id": 1051,
      "track_delete_flag": "0x00",
      "track_status": "active",
      "loss_count_cpi": 2,
      "loss_reason_code": 0,
      "update_time_cpi": 113840,
      "age_pulses": 2
    },
    "classification": {
      "track_type_code": "0x20",
      "track_type_label": "drone",
      "attributes_bitfield": "0x01",
      "radar_frontend_id": 1,
      "confidence": 0.74,
      "confidence_pct": "74%"
    },
    "signal_quality": {
      "energy": 1921.05,
      "energy_normalized": 0.62,
      "credit_ratio": 0.78,
      "credit_ratio_pct": "78%"
    },
    "polar_coordinates": {
      "distance_m": 2781.2,
      "azimuth_deg": 138.7,
      "elevation_pitch_deg": 3.9,
      "height_m": 189.4,
      "radial_speed_ms": 5.1,
      "radial_speed_label": "receding"
    },
    "geodetic_coordinates": {
      "longitude": 107.635817,
      "latitude": -6.921089,
      "altitude_m": 190.0,
      "distance_from_radar_m": 2781.2,
      "bearing_from_radar_deg": 138.7,
      "altitude_above_radar_m": 40.0
    },
    "velocity_enu": {
      "vx_east_ms": 4.2,
      "vy_north_ms": 2.8,
      "vz_up_ms": -0.6,
      "total_speed_ms": 5.05,
      "horizontal_speed_ms": 5.02,
      "vertical_direction": "descending"
    },
    "track_history": {
      "track_point_count": 14,
      "envelope_point_count": 4,
      "track_age_pulses": 14,
      "track_confidence_accumulated": true
    },
    "utc_timestamp": {
      "year": 2025,
      "month": 12,
      "day": 17,
      "hour": 8,
      "minute": 34,

```

```
    "second": 51,  
    "millisecond": 820,  
    "millisecond_precision_10ms": 82,  
    "iso8601": "2025-12-17T08:34:51.820Z",  
    "unix_timestamp": 1766049291.82  
  }  
]  
}
```

6.1 Comprehensive JSON Field Reference

This table documents EVERY field in the comprehensive JSON output, including fields from the Status Information Frame (0xFF03) that provide radar device/system data, not just target tracking data. This is the COMPLETE dataset available from the protocol.

JSON Field Path	Data Type	Source Frame	Units	Purpose & Usage
frame_metadata.frame_header	hex string	Universal frame header	–	Always 0x55AA. Frame synchronization.
frame_metadata.frame_id	hex string	Frame ID field	–	Identifies frame type (0xFF02=search, 0xFF01=track, 0xFF03=status).
frame_metadata.frame_type	string	Derived	–	Human-readable frame type name.
frame_metadata.frame_count	integer	frame_count field	counts	Monotonic counter. Detect dropped frames.
frame_metadata.frame_timestamp_utc	ISO 8601	UTC fields	string	Complete frame transmission timestamp.
protocol.id	integer	protocol_id	–	Radar device ID (1–8). Multi-radar systems.
protocol.device_code	string	Derived from ID	–	RD01–RD08 format.
protocol.major_version	integer	protocol_major	–	Major version: forward-compat only.
protocol.minor_version	integer	protocol_minor	–	Minor version: full compat within major.
protocol.checksum_valid	boolean	Derived	–	Verified frame integrity.
radar_device.device_id	integer	frontend_count_id	–	Radar unit ID. Fixed at 1 for single-head.
radar_device.device_label	string	Status frame	–	Human-readable device name/label.
radar_device.model_code	hex	radar_model field	–	0x01–0x10 → RD01–RD10.
radar_device.model_name	string	Derived	–	RD01, RD02, etc.
radar_device.network.ip_address	string	Status frame	IPv4	Radar unit IP address.
radar_device.network.port	integer	Status frame	port	Radar UDP listening port (7000).
radar_device.network.status	string	Status: frontend1_net_statuses	–	Connected/disconnected state.
radar_device.gps.longitude	float	Status frame: longitude	degrees WGS-84	Radar installation longitude (6 decimals).
radar_device.gps.latitude	float	Status frame: latitude	degrees WGS-84	Radar installation latitude (6 decimals).
radar_device.gps.elevation_m	float	Status frame: elevation	meters	Radar antenna height above sea level.
radar_device.gps.heading_angle_deg	float	Status frame: heading_angle	degrees [0,360]	Geographic azimuth radar points (from GPS). For north correction.
radar_device.gps.heading_valid	boolean	Status frame: heading_valid	–	Is GPS heading locked and trustworthy?
radar_device.gps.satellite_count	integer	Status frame: satellite_count	sats	Number of GPS satellites locked. ≥8 preferred.
radar_device.gps.installed_location_name	string	Manual/config	–	Optional: friendly location name.
radar_device.orientation.roll_angle_deg	float	Status frame: roll_angle	degrees	Inclinometer X-axis. Left-low/right-high = positive.

radar_device.orientation.pitch_angle_deg	float	Status frame: pitch_angle	degrees	Inclinometer Y-axis. Front-low/back-high = positive.
radar_device.orientation.north_correction_angle_deg	float	Status frame: north_angle	degrees [0,360]	Calculated north correction. Use directly in Standby mode.
radar_device.orientation.inclinometer_z	float	Status frame: inclinometer_z	–	Z-axis reading (reference only).
radar_device.operational_state.work_mode	hex	Status frame: work_mode	–	0x00=Standby, 0x11=Search, 0x22=Track, 0x33=Leveling.
radar_device.operational_state.work_mode_label	string	Derived	–	Human-readable mode name.
radar_device.operational_state.cmd_execution_status	hex	Status frame: cmd_exec_status	–	0x00=Idle, 0x11=Executing, 0x22=Success, 0x33=Failed.
radar_device.operational_state.fault_type	hex	Status frame: fault_type	–	0x00=None, 0x01=Parse fail, 0x02=Cannot execute.
radar_device.operational_state.servo_mode	hex	Status frame: servo_mode	–	0x11=Point, 0x33=Circular, 0x77=Pull pin, 0x88=Insert pin, 0xFF=Idle.
radar_device.operational_state.scan_mode	hex	Status frame: scan_mode	–	0x01=Mechanical, 0x02=Phase, 0x03=Composite.
radar_device.servo.status_bits.*	boolean	Status frame: servo_status	–	Encoder/driver comm, timeout, validity, pin status.
radar_device.servo.current_azimuth_deg	float	Status frame: frontendl_azimuth	degrees [0,360]	Current servo azimuth angle.
radar_device.servo.current_pitch_deg	float	Status frame: frontendl_pitch	degrees [0,360]	Current servo elevation angle.
radar_device.servo.current_scan_cycle	integer	Status frame: frontendl_scan_cycle	counts	Scan cycle counter (increments per sweep).
radar_device.frequency.current_code	integer	Status frame: frontendl_frequency	code 0– 20	Frequency selection code.
radar_device.frequency.current_x_band_ghz	float	Appendix 1 lookup	GHz	X-Band frequency (9.20–9.80 GHz).
radar_device.frequency.current_ku_band_ghz	float	Appendix 1 lookup	GHz	Ku-Band frequency (15.70–16.70 GHz).
radar_device.frequency.synthesis_status	hex	Status frame: freq_synth_status	–	LO1 and LO2 lock status.
radar_device.silent_zones[].zone_number	integer	Mode command	–	Zone 1 or 2.
radar_device.silent_zones[].start_angle_deg	float	Mode command	degrees	Silent zone start azimuth.
radar_device.silent_zones[].end_angle_deg	float	Mode command	degrees	Silent zone end azimuth.
radar_device.silent_zones[].enabled	boolean	Mode command: 0x7FFF check	–	Is this zone active?
radar_device.filters.height_min_near_zone_m	float	Mode/Status frames	meters	Min height filter ≤3 km zone.
radar_device.filters.height_min_far_zone_m	float	Mode/Status frames	meters	Min height filter >3 km zone.
radar_device.filters.height_max_m	float	Mode/Status frames	meters	Remove targets at/above this height.
radar_device.filters.speed_min_ms	float	Mode/Status frames	m/s	Remove targets slower than this (×0.1).
radar_device.filters.speed_max_ms	float	Mode/Status frames	m/s	Remove targets faster than this (×0.1).

radar_device.filters.range_min_m	float	Mode/Status frames	meters	Remove targets closer than this.
radar_device.filters.range_max_m	float	Mode/Status frames	meters	Remove targets further than this.
radar_device.filters.altitude_range_mode	hex	Mode command	–	0x11=7.5km, 0x22=15km.
radar_device.autonomous_identification.enabled	boolean	Mode command	–	Auto ID for search + track, or track-only.
radar_device.autonomous_identification.max_targets	integer	Mode command	targets	1–2. Max targets for autonomous ID.
radar_device.thermals.subarray[].index	integer	Status frame	–	Subarray 1–8.
radar_device.thermals.subarray[].temperature_c	float	Status frame	°C	Subarray temperature (precision 1°C).
radar_device.thermals.subarray[5].wave_controller_version	string	Status frame	–	Sub-5: wave controller firmware version.
radar_device.thermals.signal_board_temperature_c	float	Status frame: signal_temp	°C	Signal processing board temperature.
radar_device.thermals.frequency_synthesizer_temperature_c	float	Status frame: freq_synth_temp	°C	Frequency synthesizer temperature.
radar_device.power.subarray_currents_a[].index	integer	Status frame	–	Subarray 1–8.
radar_device.power.subarray_currents_a[].current_a	float	Status frame	amperes	Subarray draw (precision ×0.1 A).
radar_device.power.signal_voltages_v[].channel	integer	Status frame	–	Voltage channel 1–3.
radar_device.power.signal_voltages_v[].voltage_v	float	Status frame	volts	Channel voltage (precision 1 V).
radar_device.system_health.fan_status.*	string	Status frame: fan_status bits	–	Fan 1–4: normal or fault.
radar_device.system_health.memory_status.*	boolean	Status frame: info_status bits	–	QDR, DDR-0, DDR-1, PCIe init status.
radar_device.system_health.ad_channels.*	boolean	Status frame: info_status bits	–	AD over-range detection per channel.
radar_device.firmware.fpga_version	string	Status frame	–	FPGA firmware version string.
radar_device.firmware.data_processing_version	string	Status frame	–	Data processing software version.
scan_header.search_azimuth_deg	float	Search frame header: search_azimuth	degrees	Current scanner azimuth during detection.
scan_header.search_elevation_deg	float	Search frame header: search_elevation	degrees	Current scanner elevation during detection.
scan_header.scan_cycle_count	integer	Search frame header: scan_cycle	counts	Scan cycle index. Increments each 360° sweep.
scan_header.pulse_group_id	integer	Search frame header: pulse_group_id	counts	Unique pulse group identifier.
scan_header.target_count	integer	Search frame header: target_count	targets	Number of target records in this frame (0–N).
targets[].track_identification.track_id	integer	Per-target: track_id	ID	Persistent target identifier. Use for track history.
targets[].track_identification.track_delete_flag	hex	Per-target: track_delete_flag	–	0x00=active, 0x01=marked for deletion.
targets[].track_identification.loss_count_cpi	integer	Per-target: loss_count	CPI units	Consecutive pulses without re-detection.
targets[].track_identification.update_time_cpi	integer	Per-target: update_time	CPI counts	Pulse at which target was last updated.
targets[].track_identification.age_pulses	integer	Derived	pulses	delta = current_pulse_id - update_time.

targets[].classification.track_type_code	hex	Per-target: track_type	–	0x00=unknown, 0x11=person, 0x12=car, 0x20=drone, etc.
targets[].classification.track_type_label	string	Derived from code	–	drone, person, car, bird, vessel, other, false.
targets[].classification.confidence	float	Per-target: confidence	ratio [0,1]	Autonomous identification confidence (precision 0.01).
targets[].signal_quality.energy	float	Per-target: energy	dB or linear	Received signal strength. Higher = stronger return.
targets[].signal_quality.credit_ratio	float	Per-target: credit_ratio	ratio [0,1]	Track quality score. Reflects filter confidence.
targets[].polar_coordinates.distance_m	float	Per-target: distance	meters	Slant range from radar to target.
targets[].polar_coordinates.azimuth_deg	float	Per-target: azimuth	degrees [0,360]	Compass bearing to target.
targets[].polar_coordinates.elevation_pitch_deg	float	Per-target: pitch	degrees	Elevation angle. Negative=below, positive=above horizon.
targets[].polar_coordinates.height_m	float	Per-target: height	meters	Height above reference datum.
targets[].polar_coordinates.radial_speed_ms	float	Per-target: speed with sign	m/s	Doppler velocity along bore-sight. Negative=approaching.
targets[].geodetic_coordinates.longitude	float	Per-target: longitude	degrees WGS-84	Target longitude (6 decimals, ~0.1 m precision).
targets[].geodetic_coordinates.latitude	float	Per-target: latitude	degrees WGS-84	Target latitude (6 decimals, ~0.1 m precision).
targets[].geodetic_coordinates.altitude_m	float	Per-target: altitude_gps	meters WGS-84	Target GPS altitude.
targets[].geodetic_coordinates.distance_from_radar_m	float	Derived	meters	Same as polar.distance. Convenience copy.
targets[].geodetic_coordinates.bearing_from_radar_deg	float	Derived	degrees	Same as polar.azimuth. Convenience copy.
targets[].geodetic_coordinates.altitude_above_radar_m	float	Derived	meters	geodetic.altitude - radar.elevation.
targets[].velocity_enu.vx_east_ms	float	Per-target: vx	m/s	Velocity pointing East. ENU frame.
targets[].velocity_enu.vy_north_ms	float	Per-target: vy	m/s	Velocity pointing North. ENU frame.
targets[].velocity_enu.vz_up_ms	float	Per-target: vz	m/s	Velocity pointing Up. Positive=ascending.
targets[].velocity_enu.total_speed_ms	float	Derived: $\sqrt{vx^2+vy^2+vz^2}$	m/s	3D speed magnitude.
targets[].velocity_enu.horizontal_speed_ms	float	Derived: $\sqrt{vx^2+vy^2}$	m/s	Ground speed (East–North components only).
targets[].track_history.track_point_count	integer	Per-target: track_point_count	points	Number of historical positions accumulated.
targets[].track_history.envelope_point_count	integer	Per-target: envelope_point_count	points	Convex hull points for size/RCS estimation.
targets[].track_history.track_age_pulses	integer	Derived	pulses	Total pulses this track has been active.
targets[].utc_timestamp.*	integers	Per-target: UTC fields	–	Year, month, day, hour, minute, second, ms.
targets[].utc_timestamp.iso8601	string	Derived from UTC	ISO 8601	Full timestamp: YYYY-MM-DDTHH:MM:SS.sssZ.

<code>targets[].utc_timestamp.unix_timestamp</code>	float	Derived	seconds since epoch	For application timestamp correlation.
---	-------	---------	---------------------------	---

6.2 Typical Usage Patterns

Display / Visualization

Use track_id to maintain persistent display objects across frames. Update polar (azimuth, pitch, distance) or geodetic (longitude, latitude) coordinates to reposition target symbols on map/scope. Color code by track_type_label. Flash or highlight high-confidence drone (0x20) targets.

Threat Assessment

Filter on track_type_label == 'drone' AND confidence >= 0.7. Apply altitude threshold: if height_m < 100 m, escalate priority. Monitor vz_up_ms < -2.0 (rapid descent). Check distance_m for early warning thresholds.

Track Management

Group frames by track_id. Calculate age: delta_pulse = current_pulse_group_id - update_time_cpi. If delta > threshold, mark as stale. If track_delete_flag == 0x01, remove from display after timeout. Use loss_count to predict imminent deletion.

Data Logging / Analysis

Timestamp each JSON record with frame_count and utc_timestamp.iso8601. Log all targets including low-confidence detections for post-hoc analysis. Archive with protocol version and radar device ID for multi-radar correlation.

Appendix 1: Frequency Point Correspondence Table

The frequency field in Mode Command Frame accepts codes 0–20. The table below lists the corresponding X-Band and Ku-Band frequencies.

#	Code	X-Band (GHz)	Ku-Band (GHz)
1	0	9.20	15.70
2	1	9.23	15.75
3	2	9.26	15.80
4	3	9.29	15.85
5	4	9.32	15.90
6	5	9.35	15.95
7	6	9.38	16.00
8	7	9.41	16.05
9	8	9.44	16.10
10	9	9.47	16.15
11	10	9.50	16.20
12	11	9.53	16.25
13	12	9.56	16.30
14	13	9.59	16.35
15	14	9.62	16.40
16	15	9.65	16.45
17	16	9.68	16.50
18	17	9.71	16.55
19	18	9.74	16.60
20	19	9.77	16.65
21	20	9.80	16.70

Appendix 2: North Angle Calculation Algorithm

In mechanical scanning standby mode, the radar autonomously calculates the north correction angle using the GPS heading and servo azimuth. The algorithm compares the current heading angle with the mean of the previous 30 seconds of heading angles. When the delta is below 0.01° , the current heading is substituted into the function below.

Function signature and source:

```
float north_angle(float servo_angle, float course_angle) {
    float north = 0.0;
    float angle = 0.0;

    angle = course_angle - 90;

    while (angle > 360)
        angle = 360;
    while (angle < 0)
        angle += 360;

    north = servo_angle - angle;

    while (north > 360)
        north -= 360;
    while (north < 0)
        north += 360;

    return north;
}
```

Where:

Parameter	Source	Description
servo_angle	Status frame (0xFF03): frontend1_azimuth	Current mechanical servo azimuth angle (degrees)
course_angle	Status frame (0xFF03): heading_angle	GPS heading angle – geographical azimuth that the radar array normal points toward (degrees)
return value	Result → north_angle in status frame	Calculated north correction angle in degrees [0, 360]

■ **Note:** The returned north_angle value in the Status Information Frame (0xFF03) can be used directly by the operator in Standby mode.